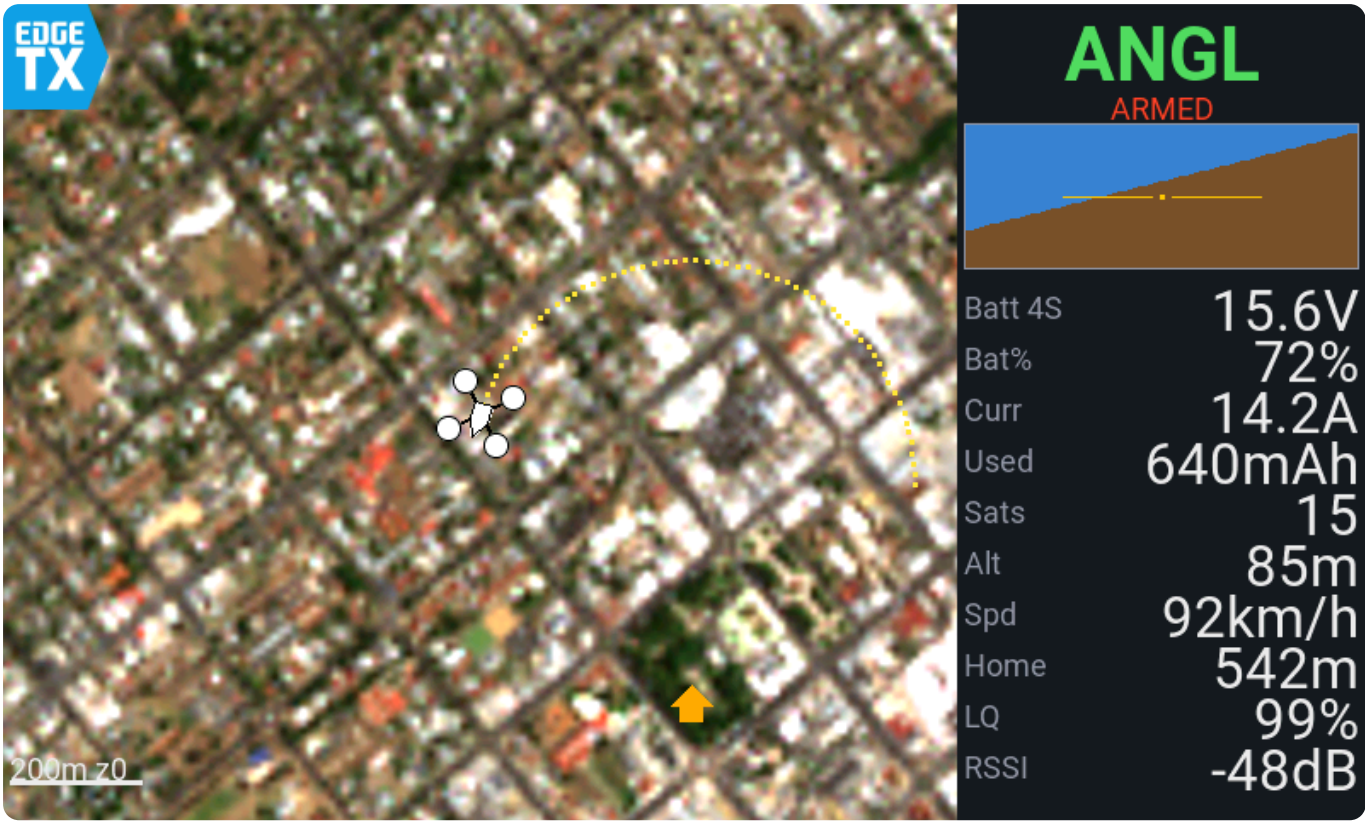
NVFPV · EDGETX LUA WIDGET

V2.1 MAVLink and mLRS

New View Telemetry

Live satellite map and flight telemetry for EdgeTX colour radios.
Installation and user guide · Version 2.1 · October 2026

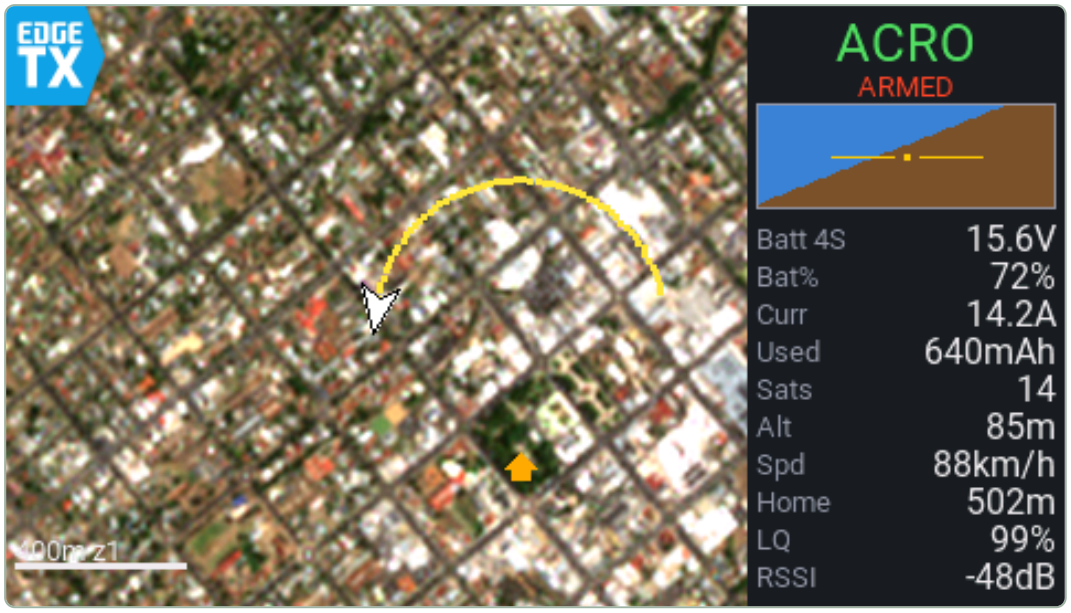


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1. What New View Telemetry does

New View Telemetry is a full-screen widget for EdgeTX colour radios. The left side is a moving satellite map of your flying area, with your craft, its trail and home. The right side is a live flight panel. Everything comes from the telemetry your receiver sends back, and the map pictures are stored on the radio's SD card, so it works in the field with no internet.



Flying, on a HelloRadioSky V15 (480 × 272 screen). Map on the left, flight panel on the right.

- **Satellite map** that follows the craft, with an arrow, drone or plane cursor pointing where the nose points.
- **Home marker** and a **yellow trail** of the last part of the flight. Home is set when you arm, and a switch can move it.
- **Zoom** from a switch, the scroll wheel or the +/- keys, with a scale bar.
- **Flight panel:** flight mode, armed state, artificial horizon, battery voltage and cell count, battery %, current, mAh used, satellites, altitude, speed, distance home, link quality and RSSI.
- **Warning limits** you set yourself. Values turn amber near a limit and red past it, with an optional beep and vibration.
- **Night-vision load screen** while it waits for telemetry and the first GPS fix, with a live satellite count.

- **Fits any colour screen:** 320×240, 480×272, 480×320, 800×480 and 320×480 portrait. The same file works on all of them.

2. What you need

Item	Details
Radio	An EdgeTX radio with a colour screen, running **EdgeTX 2.11 or later**. The HelloRadioSky V15 needs **EdgeTX 2.12** or later. Black-and-white screen radios can't run colour widgets.
Radio link	ExpressLRS or TBS Crossfire (CRSF telemetry), with telemetry turned on. Keep the transmitter module and receiver on the same ExpressLRS major version.
Flight controller	Any flight controller that sends GPS over CRSF telemetry, for example Betaflight with a GPS module. A barometer gives better altitude.
A computer	Windows PC to make map tiles (GMapCatcher and the tile maker are for Windows; the tile maker also runs on Mac and Linux with Python). Plus a way to copy files to the radio's SD card.

Common radios by screen size: **480×272** HelloRadioSky V15 and V16, RadioMaster TX16S and TX16S MkII, Jumper T16 and T18, FrSky X10 and X12S, iFlight Commando 14. **480×320** RadioMaster TX15 and GX15, Jumper T15 and T22, Flysky PL18 and ST16. **800×480** RadioMaster TX16S MK3. **320×240** HelloRadioSky V12, Flysky PA01. **320×480** Flysky NV14, EL18 and NB4+.

3. What's in this pack

```
V2.1 MAVLink and mLRS New View Telemetry/
├─ START HERE.txt                    quick install steps
├─ INSTRUCTIONS - STEP BY STEP.txt   the full steps as text
├─ MLRS AND MAVLINK SETUP.txt        mLRS and MAVLink setup
├─ New View Telemetry - User Guide.pdf   this guide
├─ LICENSE.md                        licence (CC BY-NC 4.0) and credits
├─ TESTER FEEDBACK FORM.txt          tell us how it went
├─ SD card/                          copy the contents onto the radio's SD card
│  ├─ WIDGETS/NewView/main.lua       the widget
│  ├─ WIDGETS/NewView/splash/        load screen pictures, one per screen size
│  └─ IMAGES/NewView/sat_tiles/      empty: your map tiles go here
├─ PC tools/
│  ├─ GMapCatcher/                   map downloader for Windows (double-click maps.exe)
│  ├─ make_map_tiles.bat / .py       tile maker using free satellite imagery
│  ├─ HOW TO MAKE MAP TILES.txt      tile format and step-by-step instructions
│  └─ LICENSE.md                     licence for the PC tools
├─ screenshots/                      the pictures in this guide
└─ video/                            the video tutorial
```

> **No map tiles are included.** Map tiles are pictures of one area, so you make them for the places you fly (section 5). Without tiles the widget still works: it shows "no tiles here" on a grey map, with the cursor, home and trail still drawn.

4. Install on the radio

1. Update the radio to EdgeTX 2.11 or later (2.12 or later for the V15). EdgeTX Buddy or Companion can do this.
2. Turn the radio off. Take out the SD card and put it in your computer, or connect the radio by USB and choose **USB Storage (SD)**.
3. Open the pack's `SD card` folder. Select `WIDGETS` and `IMAGES` and copy them onto the **root** of the radio's SD card. Your computer will say the folders already exist: choose to merge them. Nothing on the card is replaced except an older New View Telemetry.
4. Check the card has `/WIDGETS/NewView/main.lua` and `/WIDGETS/NewView/splash/`.
5. Eject the card safely and put it back in the radio.

> Keep the folder name exactly **NewView**, with that spelling and capitals. The widget looks for its pictures and tiles in folders with that name.

5. Make map tiles for your flying area

The radio can't download maps, so the map pictures (tiles) must be on the SD card. The pack has two ways to make them. Both make the exact format the widget needs, so you don't have to rename or resize anything.

What the radio needs	Details
Picture format	PNG, exactly 100 × 100 pixels, named `s_<number>.png`
Folder	`/IMAGES/NewView/sat_tiles/`, in the GMapCatcher folder layout: `<zoom>/<x÷1024>/<x mod 1024>/<y÷1024>/s_<y mod 1024>.png`
Zoom levels	Level 0 is the most detailed. Each level up shows twice the ground. Make the levels you set in Min zoom and Max zoom (0 to 2 by default).

Level	Scale bar	Ground per screen pixel	Good for	Tiles for 10 km x 10 km
0	200 m	about 2.6 m	close in around the field	about 1,500
1	400 m	about 5 m	normal flying	about 400
2	800 m	about 10 m	longer flights	about 100
3	1.6 km	about 20 m	long range	about 25
4	3.2 km	about 41 m	long range	under 10

Long-range pilots: set Max zoom to 3 or 4 and make those levels for a bigger area, for example 40 km x 40 km at levels 2 to 4. Freestyle pilots: levels 0 and 1 over a few km are plenty.

Option A: GMapCatcher (map viewer and downloader)

1. Open `PC tools\GMapCatcher` and double-click **maps.exe**. Nothing needs installing. If it doesn't start, run **vcredist_x86.exe** in the same folder once, then try again.
2. Type your field's name or address in the search box and press Enter, or drag the map to it.
3. Choose the **Satellite** map type.
4. Right-click the map and choose **Batch Download**. Set the area around your field (for example 5 km each way) and the zoom levels (for example 0 to 2; GMapCatcher numbers zoom levels the same way as the widget). Start it and wait until it finishes.
5. Open `%USERPROFILE%\.GMapCatcher\100x100\sat_tiles` by pasting it into the File Explorer address bar. These are the 100 × 100 tiles for the radio. (The 256 px tiles in `.GMapCatcher\sat_tiles` are for GMapCatcher itself; don't use those.)
6. Copy everything inside that folder into `/IMAGES/NewView/sat_tiles/` on the SD card, and let it merge.

> GMapCatcher downloads from online map providers. Check the provider's terms first: many don't allow bulk downloading.

Option B: the tile maker (free satellite imagery)

make_map_tiles.bat downloads free Copernicus Sentinel-2 satellite images and cuts them into tiles. The imagery is 10 m per pixel: roads, paddocks, tracks and rivers are clear, single houses are blurry at level 0.

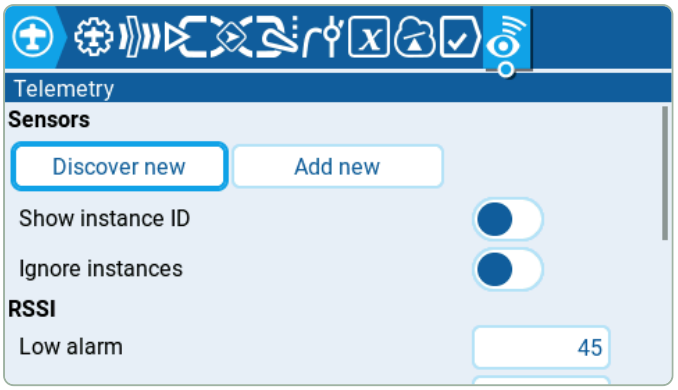
1. Install Python 3 from **python.org/downloads** and tick **Add python.exe to PATH** on the first screen.
2. In Google Maps, right-click your field and click the numbers at the top of the menu to copy them, for example `-33.8688, 151.2093`. The first number is the latitude, the second the longitude.
3. Double-click **make_map_tiles.bat** and answer the questions: latitude, longitude, the size of the square in km, and the zoom levels (press Enter for 0 1 2). The first run installs the Python packages it needs, which takes a few minutes.
4. It makes an `SD` folder next to the .bat file. Copy the `IMAGES` folder from it onto the root of the SD card and let it merge.

The full tile instructions are also in `PC tools\HOW TO MAKE MAP TILES.txt`.

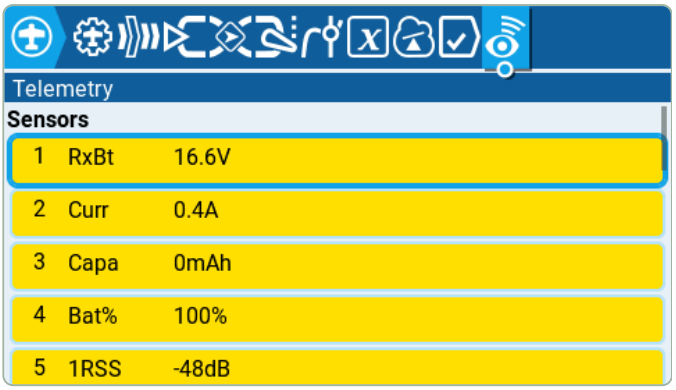
6. Find the telemetry sensors

The widget reads the standard CRSF sensors. EdgeTX has to discover them once for each model.

1. Power the craft (props off) with the GPS connected, and make sure the radio shows a telemetry link.
2. Tap the **EdgeTX** logo at the top left of the screen (or press the model key) to open the menu, then choose **Model Settings**.
3. Open the **Telemetry** tab (the last icon along the top).
4. Tap **Discover new**. Wait until the list stops growing (about 10 seconds), then tap **Stop**.



The Telemetry tab before discovery.



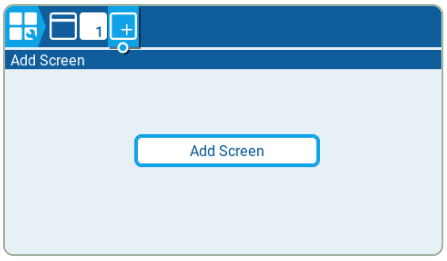
The sensors found. Rows flash as new values arrive.

Sensor	Used for	Sensor	Used for
GPS	map position, distance home, trail	FM	flight mode and armed state
Sats	satellite count	Alt (or GAlt)	altitude above home
GSpd	ground speed	Yaw / Hdg	cursor direction
Ptch, Roll	artificial horizon	RxBt, Bat%	battery voltage and %
Curr, Capa	current and mAh used	RQly, 1RSS	link quality and RSSI

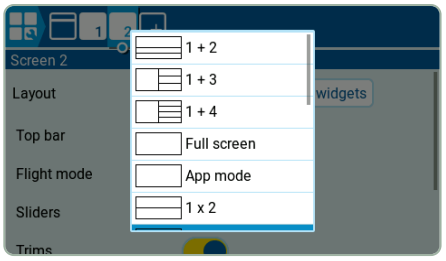
Keep the sensor names as EdgeTX makes them. If a sensor is missing, its row shows 0 and the rest still works.

7. Add the widget to a screen

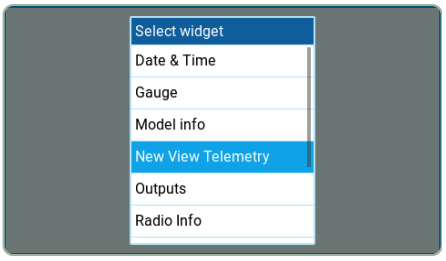
1. Select the model you fly. Tap the **EdgeTX** logo to open the menu and choose **UI Setup**.
2. Tap the **+** tab, then **Add Screen**.
3. Tap the **Layout** box and choose **App mode**, so the widget uses the whole screen (recommended). **Full screen** also works and keeps the top bar.
4. Tap **Setup widgets**, tap the empty zone, and pick **New View Telemetry** from the list.
5. The widget settings open straight away (section 8). Later you can change them by long-pressing the widget, or from Setup widgets.



UI Setup: the + tab adds a screen.



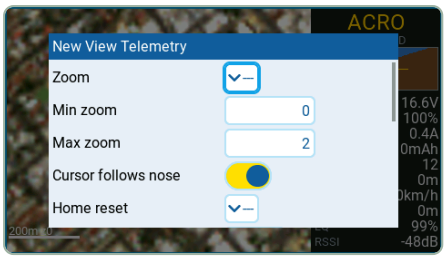
Choose App mode (or Full screen).



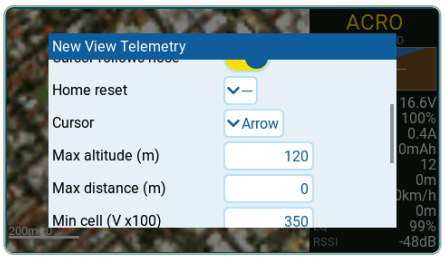
Pick New View Telemetry.

> The widget also works in a smaller zone of another layout. It works out its layout from the size it's given, but the map and panel are much easier to read full screen.

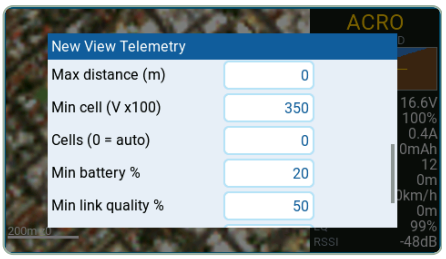
8. Widget settings



The top of Widget settings.



Cursor and the first limits.



More limits further down.

For a switch or channel setting (Zoom, Home reset), tap the box and pick it from the source list. Pick a 3-position switch for Zoom and a 2-position or momentary switch for Home reset.

Setting	Default	What it does
Zoom	none	A 3-position switch or a channel. Flick up to zoom in, down to zoom out; one step per flick, and holding it repeats. Without a zoom source, the scroll wheel or +/- keys zoom while the widget is full screen.
Min zoom	0	The most detailed zoom level allowed.
Max zoom	2	The widest zoom level allowed. Set it to the highest level you made tiles for.
Cursor follows nose	On	On: the cursor points where the nose points (Yaw). Off: it uses the heading, then the GPS track.
Home reset	none	A switch or channel. Over 50% moves home to where the craft is now.
Cursor	Arrow	The map cursor: Arrow, Drone or Plane.
Max altitude (m)	120	Altitude above home that turns red. Amber in the last 10%.
Max distance (m)	0 (off)	Distance from home that turns red.
Min cell (V x100)	350	Lowest voltage per cell, in hundredths of a volt: 350 = 3.50 V. Amber within 0.10 V.
Cells (0 = auto)	0	Battery cell count. 0 works it out from the first voltage and shows it as "Batt 4S".
Min battery %	20	Lowest battery % before it turns red.
Min link quality %	50	Lowest link quality before it turns red.
Min satellites	6	Fewest GPS satellites before it turns red.
Beep on warnings	On	Beeps and vibrates every 5 seconds while armed and past any limit.

Set any limit to **0** to turn it off. Settings are saved with each model.

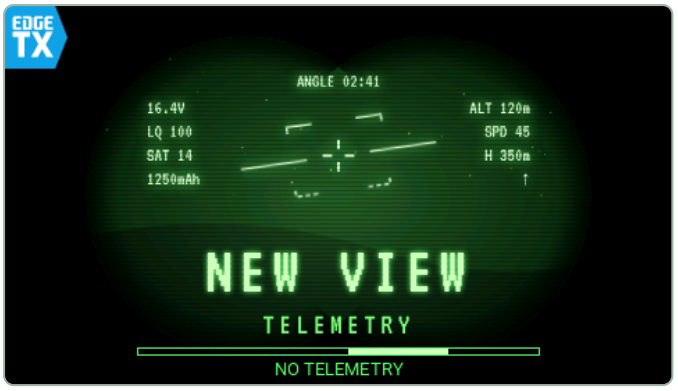
Suggested setups

Setting	Freestyle / park flying	Long range
Min zoom / Max zoom	0 / 1	1 / 4
Cursor	Drone	Plane (wing) or Drone
Max altitude	120 (or your local legal limit)	your local legal limit
Max distance	0 (off) or 500	your planned range, e.g. 5000
Min cell	350	330 for Li-ion packs
Min link quality	50	40

9. Using it in flight

Before the GPS fix

Until the first GPS fix the widget shows the night-vision load screen. The line under the title tells you what it's waiting for:



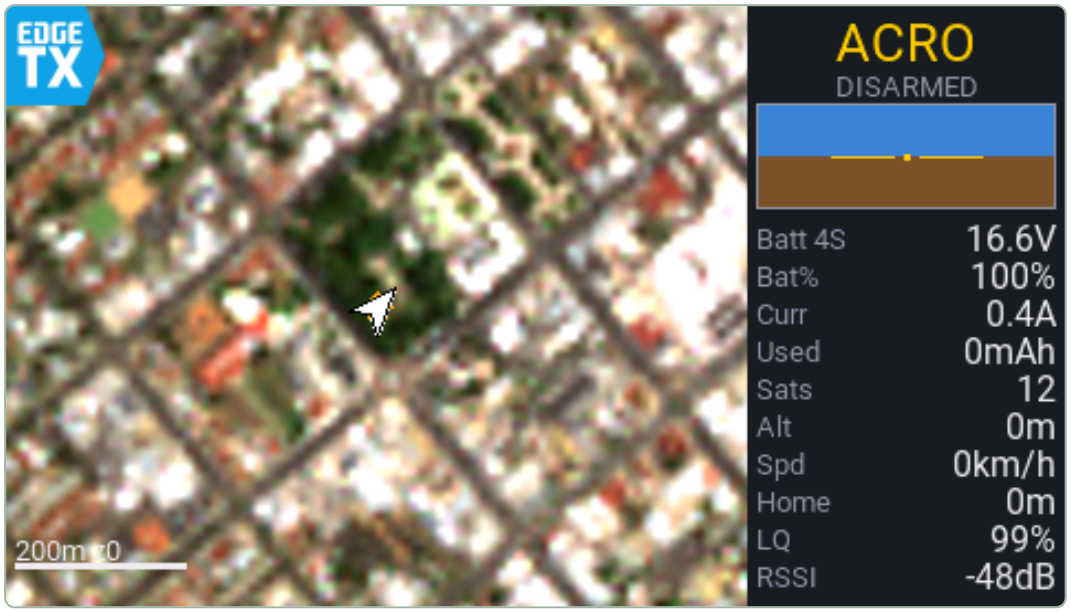
NO TELEMETRY: nothing is coming from the receiver. Power the craft and check the link.



WAITING FOR GPS - SATS 5: telemetry is working; the GPS is still getting its fix.

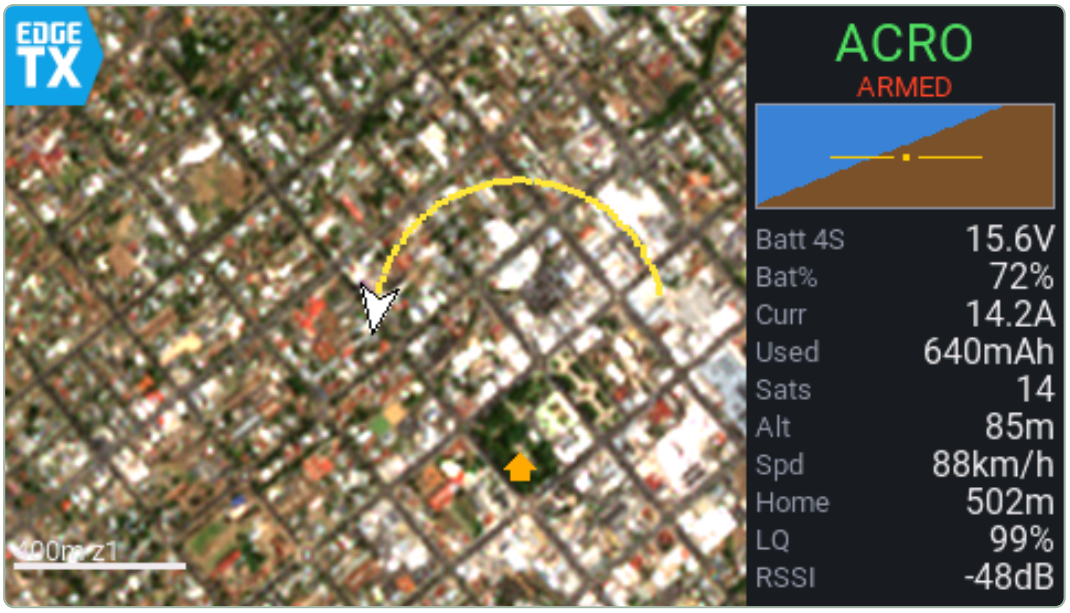
At the first fix it switches to the map. Home is set there, then again each time you arm. Wait for enough satellites before you take off, so home is accurate.

The map and the flight panel



Disarmed and ready: GPS fixed, home set under the cursor.

On screen	Meaning
Cursor (centre of map)	The craft. It points where the nose points.
Orange house	Home: where you armed, or where you last used Home reset.
Yellow dots	The trail: the last few km of the flight, one dot every 15 m. It clears when you arm.
Scale bar (bottom left)	The bar's length on the ground, and the zoom level (z0, z1...).
Flight mode (top right)	Green when armed, amber when disarmed, red in failsafe or with no telemetry. Under it: ARMED, DISARMED, ARM BLOCKED or NO GPS RESCUE.
Horizon	Pitch and roll from the flight controller.
Batt, Bat%, Curr, Used	Pack voltage (with cell count), battery %, current and mAh used.
Sats, Alt, Spd, Home	Satellites, altitude above home, ground speed and distance home.
LQ, RSSI	Link quality and signal strength.



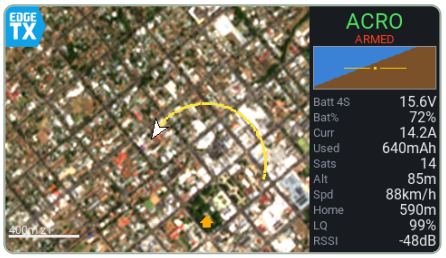
Armed and flying in Cruise mode, with the trail and home behind.

Zoom

Flick the zoom switch up to zoom in and down to zoom out, or use the scroll wheel or +/- keys when the widget is full screen. Zoom stays between Min zoom and Max zoom.



Level 0 (200 m scale bar)



Level 1 (400 m)

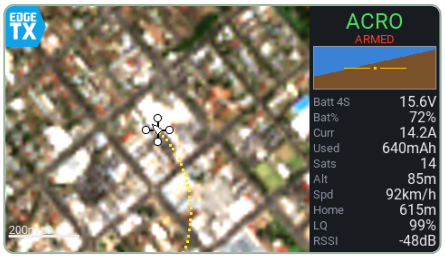


Level 2 (800 m)

Cursor choices



Arrow



Drone

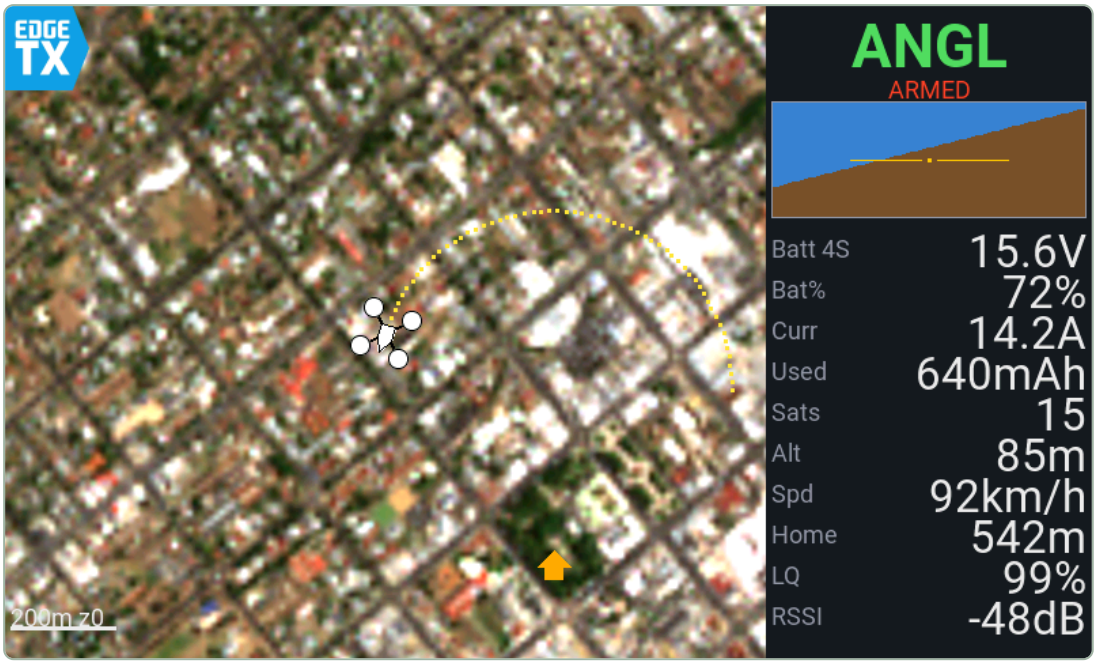


Plane

Home reset

If you armed somewhere other than where you stand, flip the Home reset switch while the craft is next to you. Home moves to the craft's position and the Home distance goes back to 0.

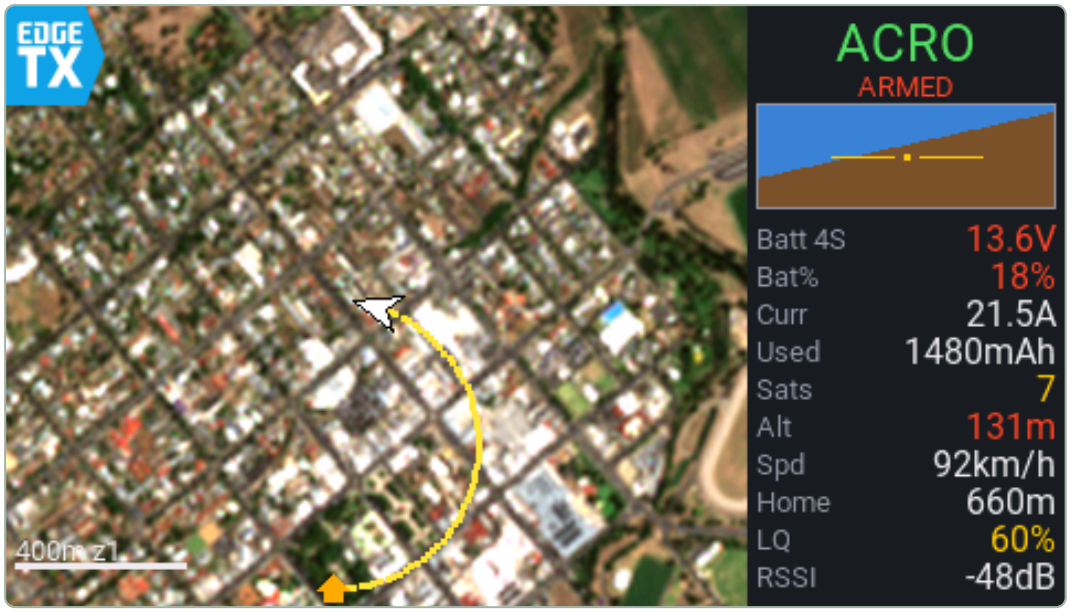
Other screen sizes



RadioMaster TX16S MK3 (800 × 480). Bigger screens show more ground and larger text.

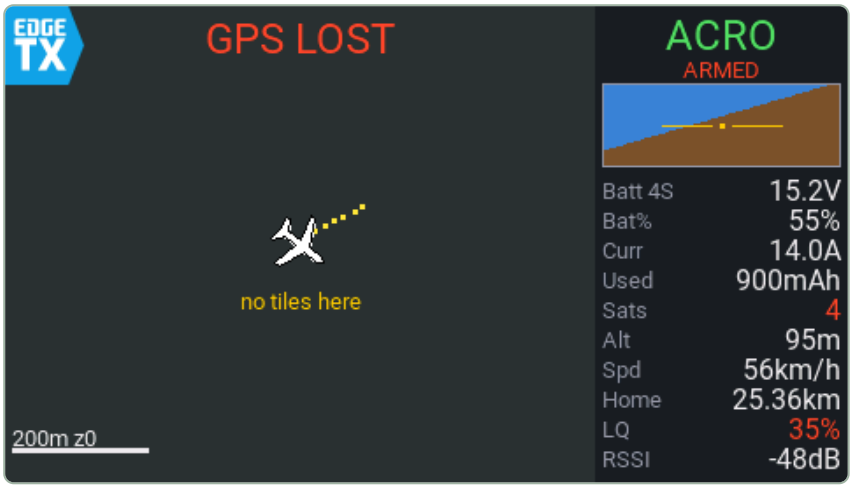
10. Warnings and limits

Each value in the panel turns **amber** close to its limit and **red** once it's past it. With Beep on warnings on, the radio also beeps and vibrates every 5 seconds while you're armed and any value is red.



Battery low (red), altitude past the 120 m limit (red), link quality close to its limit (amber).

Message	Meaning	What to do
GPS LOST (blinking red)	The GPS position hasn't updated for 3 seconds. The map shows the last position.	Check the link and GPS. Fly towards home by sight or with the OSD.
FAILSAFE	The flight controller reports failsafe.	Home and the trail stay put, so the last position on the map is where to look.
NO TELEM	No flight mode is coming in.	Check the receiver is bound and telemetry is on.
no tiles here	There are no tiles for this spot at this zoom level.	Zoom to a level you made, or make tiles for this area.



GPS LOST, flying outside the area the tiles cover.

11. Troubleshooting

Problem	Fix
New View Telemetry isn't in the widget list	Check the file is at `/WIDGETS/NewView/main.lua` (not inside an extra folder) and the radio runs EdgeTX 2.11 or later. Restart the radio after copying.
The widget shows a Lua error	Update EdgeTX, copy the pack's WIDGETS folder again, and restart the radio.
Stuck on NO TELEMETRY	Telemetry isn't reaching the radio. Check the receiver is bound, telemetry ratio isn't Off in the ExpressLRS settings, and the model's Telemetry page shows sensors updating.
Stuck on WAITING FOR GPS	The GPS has no fix yet. Go outside with a clear sky and wait. If SATS stays at 0, check the GPS is enabled in the flight controller and wired correctly.
Grey map with "no tiles here"	No tiles for this spot at this zoom. Make tiles for this area and zoom level, and check they are in `/IMAGES/NewView/sat_tiles/0`, `/1` ... on the card.
Dark squares on the map	Those tiles are in the wrong folder or aren't PNGs. Make them again with one of the PC tools.
Tiles don't line up	The tiles aren't 100 × 100 px. Use the 100×100 folder from GMapCatcher, not the 256 px one.
Zoom does nothing	Set the Zoom source in Widget settings, and check Min zoom is lower than Max zoom.
Cursor points the wrong way	Turn Cursor follows nose off to use the heading or GPS track instead.
Altitude is wrong	Without a barometer the widget uses GPS altitude minus its value at home, which is less accurate.
No beeps	Check Beep on warnings is on, the radio's volume is up, and the craft is armed. Beeps only sound while armed.
Cell count is wrong	Auto works it out from the first voltage, which can be wrong with a part-flat pack. Set Cells to your pack's cell count.

12. Credits and licences

- **V2.1 MAVLink and mLRS New View Telemetry** © 2026 New View Drone Solutions (NVFPV). Licensed CC BY-NC 4.0: free for non-commercial use, with credit. See `LICENSE.md`.
- **GMapCatcher** is free software under the GNU GPL version 2 by the GMapCatcher team, not by NVFPV. This copy is the 100×100 tile fork by Alessandro Apostoli (yaapu). Licence and source code links: `PC tools\GMapCatcher\ABOUT-THIS-COPY.txt`. Keep those files with it if you share it.
- Tile maker imagery contains modified Copernicus Sentinel data (ESA); GMapCatcher tiles follow each map provider's terms. Screenshots: EdgeTX simulator, simulated telemetry.

Fly safely and follow your local drone rules. New View Telemetry is a helper, not a replacement for keeping your aircraft in sight where the law requires it.